Topic 2 — Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र)

Covers syllabus (click to expand)

Ecology – Meaning | Ecosystem | Types of Ecosystems | Terrestrial Ecosystem | Aquatic Ecosystem | Structure and Functions of Ecosystem | Biotic and Abiotic Components (अजैविक घटक) | Ecological Niche (पारिस्थितिक निकेत) | Ecological Principles | Ecological Balance | Ecological Succession (उत्तराधिकार क्रम) | Ecosystem Services | Ecosystem Productivity | Ecological Efficiency | Genetic Level Ecology | Species Level Ecology | Community Level Ecology

Sources baked in: NCERT Class 11–12 Biology, Odum, MEA 2005, TEEB/IPBES, UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ Very High — core Prelims conceptual topic **Last verified:** August 2026

Current Affairs (this topic)

- Ecosystem-services and productivity stems stay conceptual — refresh MEA / IPBES wording if a CA stem cites (साइट्स) them.
- Keep Tansley / Haeckel / Odum name traps stable; they are not CA-dependent.

2.1 Ecology – Meaning

Identity: Study of interactions | Haeckel 1866 | 7 levels | Autoecology vs Synecology | Bionomics

- **Ecology** (from Greek *Oikos* = house + *Logos* = study) is the study of relationships between living organisms and their environment.
- **Ernst Haeckel** coined the term **Ecology** in **1866**.
- **Bionomics** (Greek *bio* = life; *nomos* = law) is a comprehensive study of an organism in relation to its environment. It is **synonymous with ecology**.
- Bionomics stresses the value of natural systems that influence human systems. It does **not** mean “management of life.”
- NCERT definition: study of relationships of organisms with each other and with their environment.
- Ecology is a **biological science** — distinct from environment studies (policy/law/economics).
- **Ramdeo Misra** is regarded as the father of **ecology in India** (ecology teaching at BHU).
- **Eugene P. Odum** is called the father of modern **ecosystem ecology** (*Fundamentals of Ecology*, 1953).
- **Arne Næss** coined **deep ecology** in **1973**.
- **John W. Bennett** used the concept of **Ecological Transition** in *The Ecological Transition: Cultural Anthropology and Human Adaptation*.

- Ecology is studied at **seven levels**: Organism → Population → Community → Ecosystem → Landscape → Biome → Biosphere.
- A shorter organisation order often asked from organism upward is **Population** → **Community** → **Ecosystem** → **Landscape**.
- **Organ** is anatomy, **not** an ecological organisation level.
- **Autoecology** studies one species in its environment.
- **Synecology** studies communities and ecosystems together.

Level	example
Organism	Camel water conservation in desert
Population	Tiger population in Dudhwa (दुधवा)
Community	Fish, birds, and plants sharing a wetland
Ecosystem	Ganga stretch at Varanasi (वाराणसी) with its biota, water, and nutrient flows
Biome	Tropical deciduous forest of India
Biosphere	Global life-support zone

> **Logic:** Trap — Odum did **not** coin (मुद्रा) "ecosystem". **Tansley (1935)** did.

2.2 Ecosystem

Identity: Tansley 1935 | Functional unit | Open system | Natural vs anthropogenic | Four components

- **A.G. Tansley** introduced the term **ecosystem** in **1935**.
- An ecosystem is the basic **functional unit** where living organisms interact with each other and the physical environment (भौतिक पर्यावरण).
- A standard short definition is a community of organisms together with the environment in which they live.
- Every ecosystem has **biotic components** (जैविक घटक), **abiotic components**, **energy flow**, and **nutrient cycling**.
- Four components frequently listed together are **abiotic materials**, **producers**, **consumers**, and **decomposers**.
- **Energy flow** is **unidirectional** (Sun → producers → consumers → heat loss).
- **Nutrient cycling** is **cyclic** — matter is reused through decomposers.
- Most ecosystems are **open systems** (exchange energy and matter with surroundings). Calling an ecosystem a **closed system** is wrong.
- The **biosphere** itself is an open system. New substances can enter, and energy can enter and leave.
- Components of an ecosystem **are interdependent**. A rise or fall in one population can reshape the rest of the food web (खाद्य जाल).
- **Natural ecosystems** self-regulate (forest, lake, ocean).

- **Anthropogenic / artificial ecosystems** depend on human inputs (cropland, rice field, orchard, garden, aquarium, sewage-treatment wetland, spaceship life-support systems).
- Among ecosystem types, the **marine (ocean) ecosystem** covers about **71%** of Earth's surface and holds about **97%** of planetary water. It is the largest and most stable ecosystem type in common option lists.
- The **biosphere** is the largest ecosystem of Earth as a whole because it integrates atmosphere, hydrosphere, and lithosphere where life exists.
- Ecosystem size runs from **micro** (drop of pond water, garden pond) to **macro** (river basin, biosphere).

> **Logic:** Ecosystem = open, not closed. Components are interdependent. Field/rice field = artificial. Biosphere = largest overall; ocean = largest/most stable type.

2.3 Types of Ecosystems

Identity: Natural/Anthropogenic | Terrestrial/Aquatic | Lentic/Lotic

Basis	Categories
Origin	Natural vs **Anthropogenic** (human-modified)
Medium	**Terrestrial** (~29% Earth's surface) vs **Aquatic** (~71%)
Water flow (freshwater)	**Lentic** (standing) vs **Lotic** (flowing)
Salinity	Freshwater vs Brackish vs Marine

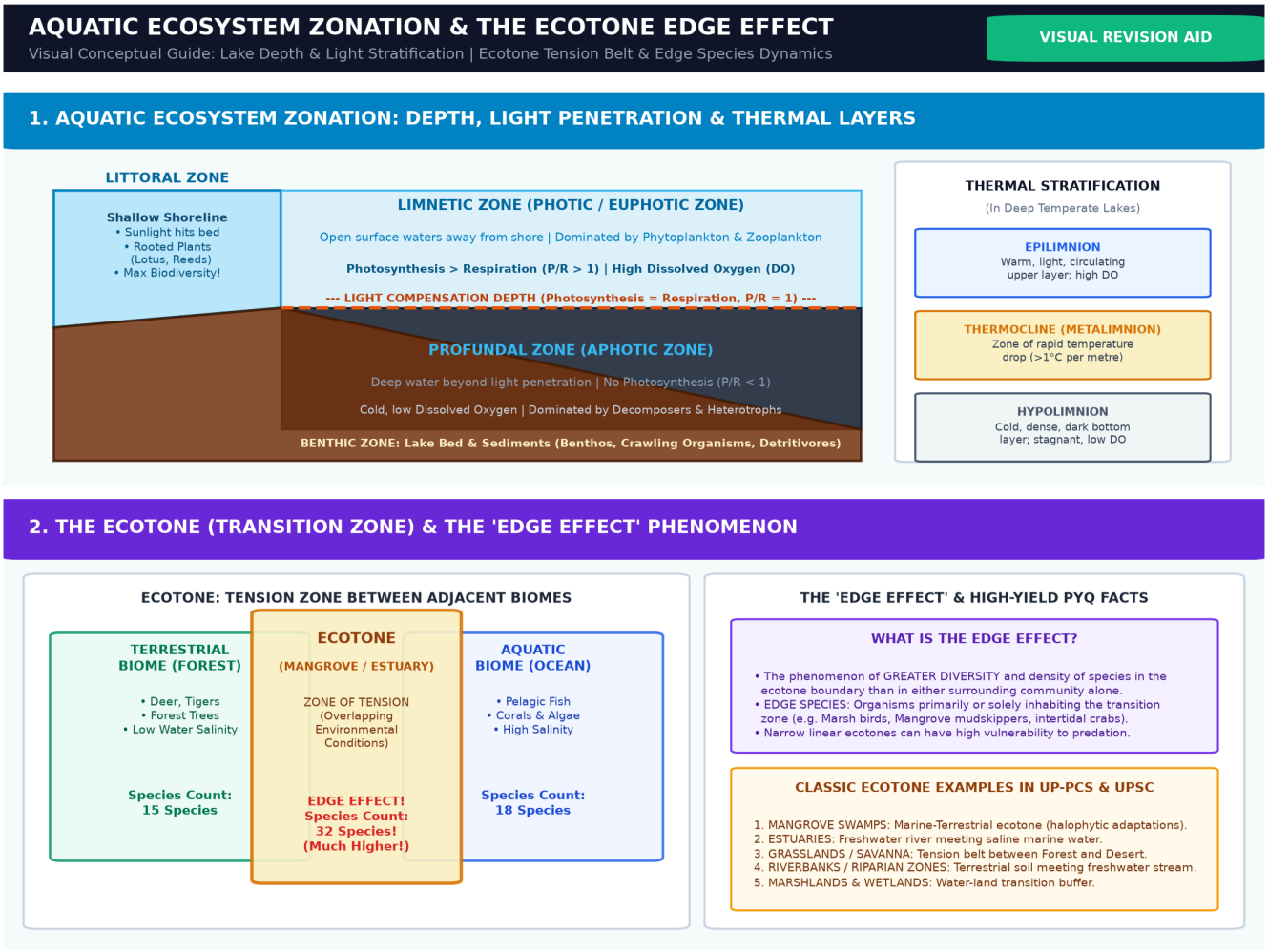
Terrestrial ecosystems

- A **terrestrial ecosystem** exists on land. Plants and animals that live on land are called **terrestrial**.
- A **forest ecosystem** is dominated by trees. The Terai sal forests and Western Ghats (पश्चिमी घाट) evergreen tracts are common Indian examples.
- A **grassland ecosystem** is dominated by grasses. Dudhwa Terai grasslands are a UP-relevant example.
- A **desert ecosystem** is water-scarce and extreme in temperature. The Thar (थार) is India's classic desert tract.
- Animals adapted to desert dryness are called **xericole**.
- A **savanna** (सवाना) mixes grassland with scattered trees. Gir (गीर) in Gujarat is India's best-known savanna-style lion (सिंह) habitat.
- **Mountain ecosystems** show vegetation change with altitude along the Himalaya (हिमालय) belts.
- **Tundra** (टुंड्रा) and **taiga** are cold-climate terrestrial types. They are not typical of the Indian mainland in Prelims options.

Aquatic ecosystems — lentic and lotic

- An **aquatic ecosystem** is water-based. **Lentic** water stands still in lakes and ponds. **Lotic** water flows in rivers and streams.

- A **lentic ecosystem** (स्थिर जल) is standing freshwater such as a lake, pond, or reservoir. Loktak and Dal Lake are lentic examples.
- A **lotic ecosystem** (प्रवाही जल) is flowing freshwater such as a river or stream. The Ganga and Yamuna are lotic.
- **Wetlands**, **estuaries**, and **mangroves** are **transitional**. They are neither purely terrestrial nor purely open aquatic.



Pond and lake zonation (lentic)

Zone	Location	Key fact
Littoral	Shallow shore with rooted plants	Warmest, highest light
Limnetic	Open water away from shore	Phytoplankton zone
Profundal	Deep water below light penetration	Low light, decomposers dominate

- The **littoral zone** is the shallow margin where rooted plants receive strong sunlight.
- The **limnetic zone** is the open-water body of a lake or pond away from the shore.
- **Phytoplankton** grow abundantly in the limnetic zone because open water receives enough light for photosynthesis.

- The **profundal zone** lies below the lighted layer. Decomposers are important here because dead organic matter sinks down.

Marine zones

- The **littoral (intertidal) zone** is the shore area alternately submerged and exposed by tides.
- The **neritic zone** covers the shallow continental shelf down to about 200 m. Fisheries and coral reefs are concentrated here.
- The **pelagic zone** is the open ocean water column away from the bottom.
- The **benthic zone** is the sea floor and its sediments.
- The **abyssal zone** is the deep ocean beyond the continental slope.
- A human-made **cropland** is an **anthropogenic biome**. It depends on irrigation, fertiliser, and harvest removal unlike a self-regulating natural forest.

> **Logic:** River = **lotic**. Lake = **lentic**. Cropland = **anthropogenic**, not natural grassland.

2.4 Structure, Functions & Components

Identity: Biotic structure | Four ecosystem functions | Nutrient cycling

Biotic components

- **Producers (autotrophs)** convert solar energy into food through photosynthesis or chemosynthesis. Green plants, algae, and cyanobacteria are the main producers.
- **Consumers (heterotrophs)** feed on other organisms. They include herbivores, carnivores, and omnivores arranged in trophic levels.
- Consumers do **not** make food by photosynthesis. That statement is false.
- **Decomposers** break down dead organic matter and return minerals to the environment. Bacteria and fungi are the classic decomposers. Viruses are **not** treated as decomposer organisms in standard food-chain questions.
- Decomposers convert **organic matter into inorganic** nutrients usable by producers. The reverse statement is false.
- **Producers and decomposers** act as intermediaries between biotic and abiotic components.
- **Detritivores** feed directly on detritus. Earthworms are a common example.
- In a typical stable ecosystem, **producers outnumber primary consumers**.
- Air is an **abiotic** component. Plants, animals, and bacteria are biotic.

Abiotic components

- **Abiotic components** are the non-living physical and chemical factors of an ecosystem.
- **Climatic factors** include temperature, rainfall, sunlight, wind, and humidity.
- **Edaphic factors** relate to soil texture, minerals, pH, and moisture.
- **Topographic factors** include altitude, slope, and aspect of the land.

- **Aquatic abiotic factors** include salinity, dissolved oxygen, and water current.
- Abiotic factors decide which species can survive in a habitat. The scarcest limiting factor (Liebig) and tolerance range (Shelford) govern which organisms persist.

Four functions of an ecosystem

Function	What it means
Productivity	Rate at which biomass or energy is produced
Decomposition	Breakdown of dead organic matter
Nutrient cycling	Movement of chemical elements through biotic and abiotic pools
Energy flow	One-way transfer of energy through trophic levels

- The movement of nutrient elements through living and non-living parts of an ecosystem is called a **bio-geochemical cycle**.
- In the **phosphorus cycle**, weathering of rocks is the main source that releases the nutrient into the cycle. Carbon and nitrogen cycles are not rock-weathering dominated in the same way.
- **Food chains**, **food webs**, and **ecological pyramids** are covered in the next syllabus topic. Here the fact is that energy flow is unidirectional through trophic levels.

Ecotone and aquatic clines

- An **ecotone** is the transition zone or boundary between two distinct communities or ecosystems.
- **Pycnocline** shows a **density** gradient in a water column.
- **Halocline** shows a **salinity** gradient.
- **Thermocline** shows temperature change with depth.
- **Marine upwelling** replaces surface water with colder, nutrient-rich deep water and raises biological productivity and fisheries.

> **Logic:** A **sere** is the community sequence in succession. It is **not** the name for nutrient cycling. Ecotone ≠ ecosphere.

2.5 Ecological Niche

Identity: Role not place | Grinnell 1917 | Fundamental vs realized | Competitive exclusion

- A **habitat** is the place where a species lives. It answers the "where" question.
- A **niche** is the functional role of a species in an ecosystem. It answers the "what job" question — resources used, conditions tolerated, and interaction with other species.
- **Joseph Grinnell (1917)** coined the ecological-niche idea and treated it as a micro-habitat / distributional unit of species.
- Each species has a **distinct niche**. The niche covers the range of conditions a species can tolerate, the resources it uses, and its functional role.

- The **fundamental niche** is the full role a species could play if no competition or predation limited it.
- The **realized niche** is the narrower role a species actually occupies after competitors, predators, or resource limits restrict it.
- **Gause's competitive exclusion principle** states that two species competing for exactly the same niche cannot coexist forever. One species is excluded, or the niche gets divided between them.
- When resources are abundant and spread fairly, **more species** can coexist in a community. UPPCS 2024 tests the false claim that species number becomes **small** when resources are sufficient.
- **Dominant species** often occupy a **broader niche** because they control more resources in the habitat.

> **Logic:** Habitat = place. Niche = role (Grinnell). More resources → more species, not fewer.

PYQ — UPPCS Prelims 2024, Q9

Which statement is **not true** about "ecological niche"?

Options: A. Dominant species occupy extensive and broader ecological niche. B. Number of species becomes large in ecological niche of natural ecosystem, if resources are sufficient. C. Number of species becomes small in ecological niche, if resources are sufficient. D. Dominance of single species becomes minimum if the resource distribution is equitable in a habitat.

Show answer

Ans: C — More resources support **more** species, not fewer.

2.6 Ecological Principles

Identity: Allen/Bergmann/Gloger | Limiting similarity | Liebig + Shelford

- **Liebig's Law of Minimum** states that growth is limited by the scarcest essential resource. In deserts, water is often that limiting factor even when sunlight is abundant.
- **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a minimum–maximum range for each factor. Beyond that range, stress or death follows even if other factors are plentiful.
- **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.
- **Allen's Rule** states that animals in colder climates tend to have smaller appendages such as ears and limbs. Smaller surface area reduces heat loss.
- **Bergmann's Rule** states that animals in colder climates tend to have larger body size. Bulk helps conserve body heat.
- **Gloger's Rule** states that darker pigmentation is more common in warm and humid regions.

> **Logic:** Allen = smaller appendages in cold. Bergmann = larger body in cold. Do not swap the pair.

2.7 Ecological Balance

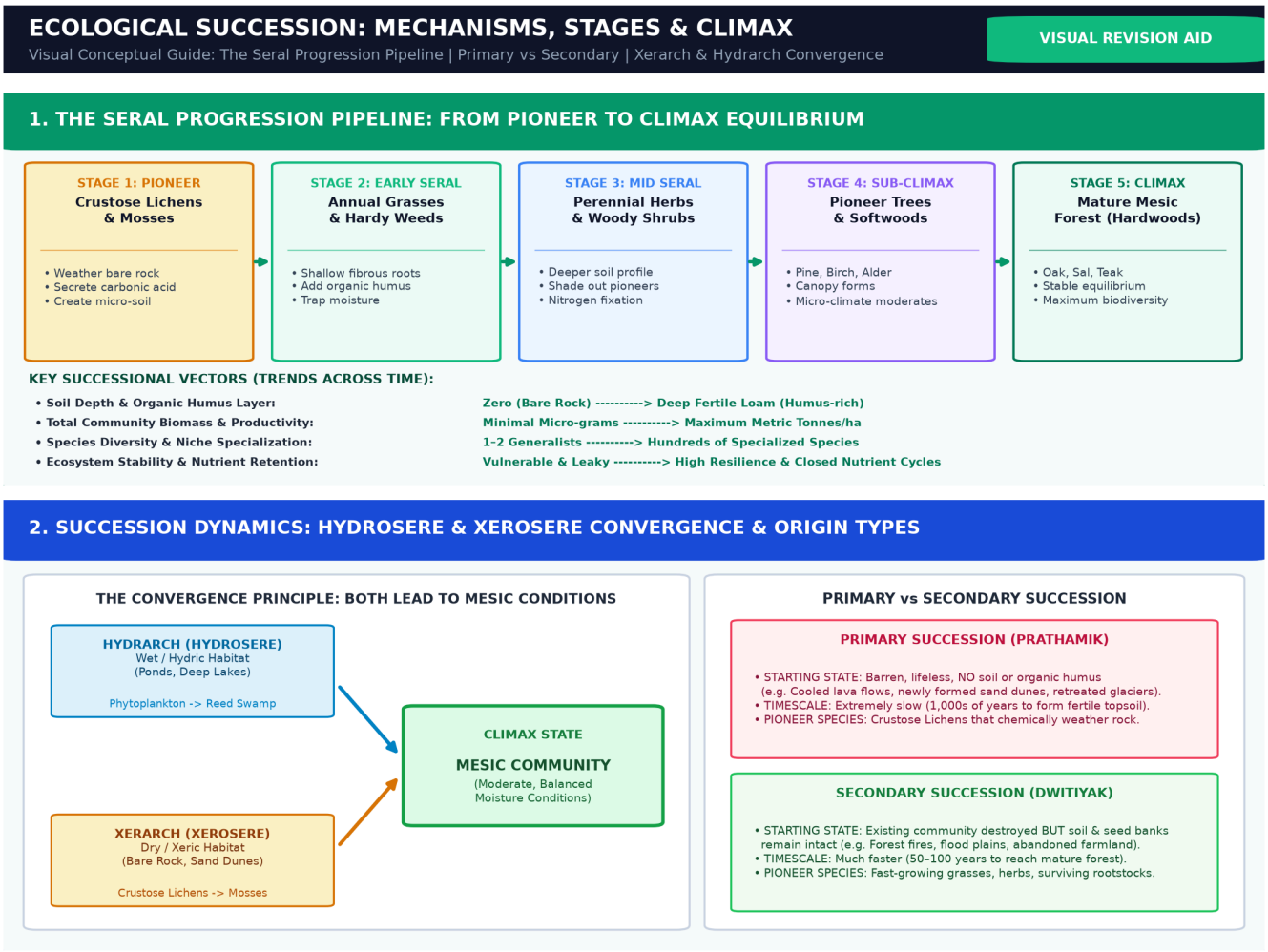
Identity: Dynamic equilibrium | Resilience | Deforestation | Eutrophication

- Ecological balance is **dynamic equilibrium**. An ecosystem looks stable but species populations constantly adjust through births, deaths, and interactions.
- **Feedback loops** such as predator–prey cycles and nutrient recycling help ecosystems absorb small disturbances without collapsing.
- **Resilience** is the ability of an ecosystem to recover after a disturbance such as fire, flood, or pollution spike.
- A **keystone species** (कीस्टोन प्रजाति) has an influence on community structure far greater than its numbers suggest. Removing it can reshape the whole ecosystem.
- Rapid human population growth **disrupts** environmental balance.
- **Deforestation** is one of the main causes of ecological imbalance in India. Desertification (मरुस्थलीकरण), floods, famines, and rainfall variation are related stresses, but forest loss is the primary driver in many stems.
- Acts that disturb balance include **lumbering** / commercial felling. Afforestation, rainwater management, biosphere reserves (जैवमंडल आरक्षित क्षेत्र), and wildlife protection (वन्यजीव सुरक्षा) help restore balance.
- Industrial management alone is **not** treated as an ecological-balance tool in standard option sets that contrast it with water management, afforestation, and wildlife protection.
- **Eucalyptus** plantations are often called an environmental hazard because the tree draws heavy groundwater and can lower the water table.

Eutrophication — imbalance in aquatic ecosystems

- **Eutrophication** begins when excess nutrients (often nitrates and phosphates from fertiliser runoff) enter a water body.
- The nutrient surge triggers rapid **algal blooms** on the surface.
- When algae die, decomposers multiply and consume **dissolved oxygen** faster than it can be replenished.
- Fish and other aerobic organisms die from oxygen starvation. marks **both** nutrient enrichment and oxygen depletion as correct.

> **Logic:** Eutrophication statement 1 (nutrient enrichment) and statement 2 (DO depletion) are **both true**.



2.8 Ecological Succession

Identity: Primary vs secondary | Nudation → M-E-R-S | Sere | Pioneer → climax | Monoclimax

- **Ecological succession** is the orderly, predictable replacement of one community by another over time in the same area.
- The full set of communities that appear one after another is called a **sere**. Each intermediate community is a **seral stage**.
- The final relatively stable community is the **climax community**.
- **F.E. Clements** described succession toward climax and is linked to the **monoclimax** theory — one climatic climax for a region.

Primary vs secondary succession

- **Primary succession** starts on bare lifeless substrate where no soil yet exists. Examples include cooled lava, bare rock, and new sand dunes.
- **Secondary succession** starts where soil and some organisms already remain after a disturbance. Examples include abandoned farmland, forest after fire, and flooded land once water recedes.
- **Pioneer species** are the first colonisers. Lichens on rock and moss on thin soil are classic terrestrial pioneers. Phytoplankton may pioneer a new pond.

Lichens as pioneer and pollution indicator

- **Lichens** often pioneer bare rock in primary succession because they tolerate harsh exposed surfaces.
- Lichens are also sensitive **bio-indicators of air pollution**. They absorb pollutants from the air and fail to grow in dirty air.
- Trap: "lichens grow well in polluted areas" is **false**.

Phases of biotic succession (full sequence)

Phase	What happens
Nudation	Creation of a bare site free of organisms (erosion, fire, glaciation, human clearance)
Migration	Seeds, spores, or other propagules reach the bare or disturbed area
Ecesis	The newcomers successfully establish, germinate, and grow (colonisation)
Reaction	Established species modify the habitat (soil, shade, moisture) for the next community
Stabilisation	Change slows and the climax community dominates

- Full Odum/Clements-style order: **Nudation** → **Migration** → **Ecesis** → **Reaction** → **Stabilisation**.
- UPPCS 2024 starts from Migration when Nudation is omitted: **Migration** → **Ecesis** → **Reaction** → **Stabilisation**.
- Trap: putting **Ecesis before Migration** is always wrong.
- In many grasslands, trees do not replace grasses in succession because of **water limits and fire**.

Hydrosere and xerosere

- A **hydrosere** is succession in an aquatic habitat that gradually becomes terrestrial. A pond may pass through floating plants, reeds, marsh, meadow, and finally forest if sediment fills it in.
- A **xerosere** is succession on dry land starting from bare rock or sand. The usual sequence runs lichen → moss → grass → shrub → forest as soil builds up.

> **Logic: Sere** = the community **sequence**. Full five phases begin with **Nudation**.

PYQ — UPPCS Prelims 2024, Q26

Correct sequence of phases of Biotic succession?

Options: A. Ecesis – Migration – Reaction – Stabilisation B. Ecesis – Migration – Stabilisation – Reaction C. Migration – Reaction – Stabilisation – Ecesis D. Migration – Ecesis – Reaction – Stabilisation

Show answer

Ans: D

2.9 Ecosystem Services

Identity: MEA 2005 | Four (sometimes five) categories | Supporting traps

Source: Millennium Ecosystem Assessment (MEA), **2005**

Category	What humans get	Example
Provisioning	Direct material products	Food, water, timber
Regulating	Processes that stabilise the environment	Climate/disease control
Supporting	Services that underpin all other services	Nutrient cycling, soil formation
Cultural	Non-material benefits	Recreation, spiritual value
Preserving (WDR/UPSC five-fold framing)	Long-term diversity maintenance	Maintenance of diversity

- **Provisioning services** supply goods humans take directly from nature. Timber from a forest is a provisioning service. Production of food and water sits here.
- **Regulating services** control natural processes that affect human welfare. Climate and disease control sit here. Bee pollination of crops is often treated as regulating, though some stems place crop pollination with supporting.
- **Supporting services** maintain the basic conditions for life. Nutrient cycling and soil formation are supporting services. In the UPSC MEA framing that lists five categories, **nutrient cycling and crop pollination** are the supporting pair tested.
- **Cultural services** are non-material benefits such as recreation at a national park or spiritual value of a sacred grove.
- **Preserving services**, when listed separately, cover maintenance of diversity over long time scales.
- **Supporting services** underlie provisioning and regulating services. Without soil formation and primary production, food and timber cannot exist.
- **TEEB** and **IPBES** later pushed economic valuation of ecosystem services for policy.

> **Logic:** Timber / food-water = **Provisioning**. Climate-disease control = **Regulating**. Nutrient cycling = **Supporting**. Maintenance of diversity = **Preserving** in five-fold lists.

2.10 Ecosystem Productivity

Identity: GPP | NPP | 1% sunlight rule | Standing crop

Term	Meaning
GPP (Gross Primary Productivity)	Total energy or biomass fixed by producers through photosynthesis

Term	Meaning
R	Energy lost by producers through respiration
NPP (Net Primary Productivity)	**GPP – R** = biomass actually available to consumers
NSP	NPP remaining after heterotroph consumption at producer level
Standing crop	Biomass present in an ecosystem at a given instant

- **Productivity** measures how fast an ecosystem produces organic matter. Standard units are **g/m²/yr** or **kcal/m²/yr**.
- Terrestrial green plants capture only about **1%** of the solar energy that falls on their leaves (NCERT standard; statement 1).
- Most incoming sunlight is reflected, transmitted through leaves, or used in ways that do not become stored biomass.
- **GPP** counts all energy fixed by photosynthesis. Plants immediately use part of it in **respiration (R)**.
- **NPP** is what remains for herbivores and other consumers after plant respiration is subtracted.
- **Tropical rainforests** and **coral reefs** rank among the highest-NPP ecosystems because they combine strong sunlight with ample moisture.
- **Deserts** show low NPP despite bright sunlight because water availability is the limiting factor.
- A decreasing productivity sequence often tested is **Mangroves → grasslands → lakes → oceans**.
- Most of the world's coral reefs lie in **tropical waters**. More than one-third are in the territories of **Australia, Indonesia, and the Philippines**.
- Coral reefs host far more animal **phyla** than tropical rainforests, despite covering a tiny share of ocean floor.

> **Logic:** The **1% rule** applies to energy **captured by producers**. The **10% rule** applies to energy **transferred between trophic levels**. Both can be true in the same question. Productivity order ≠ stability order.

2.11 Ecological Efficiency

Identity: Lindeman 10% rule | Trophic levels | Energy loss as heat

- **Raymond Lindeman (1942)** showed that only about **10%** of energy or organic matter passes from one trophic level to the next higher level.
- About **90%** is lost as heat through respiration, movement, excretion, and energy in uneaten parts of organisms.
- Because of this loss, energy **decreases** at each higher trophic level.
- **Trophic levels** are the feeding steps in a food chain (खाद्य श्रृंखला). They are formed by organisms linked in a food chain.
- **Ecological efficiency** compares energy at two trophic levels: (energy at higher level ÷ energy at lower level) × 100.

- A **pyramid (पिरामिड) of energy** (ऊर्जा पिरामिड) is always upright because energy cannot increase as you move up the chain.

PYQ — UPPCS Prelims 2025, Q5

With reference to 'Ecosystem', which statements is/are correct?

1. Green plants in terrestrial ecosystem capture about 1% of energy of the sunlight that falls on their leaves.
2. Around 10% of organic matter reaches to the next higher level of consumers.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: C — Statement 1 = 1% capture by producers. Statement 2 = 10% transfer to next trophic level. Both true together.

> **Logic:** Trap — "energy increases up the food chain" is **false**. Trap — rejecting statement 1 because statement 2 exists is **false**.

2.12 Genetic, Species & Community Ecology

Identity: Three study levels below ecosystem | Population curves | Species interactions

Genetic level ecology

- **Genetic-level ecology** studies gene pools, allele frequencies, and genetic variation within populations.
- High **genetic diversity** (आनुवंशिक विविधता) helps a population survive disease outbreaks and environmental change.
- The **Hardy–Weinberg equilibrium** is an ideal model where allele frequencies stay constant when no evolutionary forces act.

Species level ecology (population ecology)

- A **population** is a group of individuals of the same species living in one area and interbreeding.
- **Carrying capacity (K)** is the maximum population size an environment can support indefinitely without degradation, given food, habitat, water, and other necessities.
- **Biotic potential (r)** is the maximum reproductive capacity of an organism under optimum conditions. Carrying capacity is the environmental limit beyond which that growth cannot be sustained.

- Exceeding carrying capacity stresses the ecosystem and can cause collapse.
- A **J-curve** shows exponential population growth when resources seem unlimited at first.
- An **S-curve** shows logistic growth slowing as the population approaches carrying capacity K.
- An **r-strategist** species produces many offspring quickly and invests little in each individual. Mosquitoes and weeds are examples.
- A **K-strategist** species produces few offspring and invests heavily in parental care. Elephants, whales, and tigers are examples.

Community level ecology

- A **community** includes all populations of different species living together in one area.
- **Species richness** counts how many species are present. **Evenness** measures how evenly individuals are spread among those species.
- Among common options, **coral reefs** show relatively **highest species diversity (प्रजाति विविधता)**.
- The **marine (ocean) ecosystem** is often cited as the **most stable** ecosystem type. Do not swap diversity with stability.
- **Habitat destruction** is the leading direct cause of biodiversity loss.

Interaction	What happens	Example
Competition	Two species vie for the same limited resource	Two grazers on one pasture
Predation	One organism kills and eats another	Tiger hunting deer
Parasitism	One benefits while the host is harmed	Tick on cattle
Mutualism	Both species benefit	**Azolla** with *Anabaena* in rice fields
Commensalism	One benefits and the other is unaffected	Cattle egret feeding near grazing cattle

> **Logic:** Coral reefs = high **diversity**. Open ocean = high **stability**. These are different 2018 questions.

2.13 Ecological Succession, Seral Stages & Biogeochemical Dynamics

- **Ecological Succession**: The universal and predictable process by which an ecological community undergoes orderly and gradual changes in species composition over time until a stable state is established.
- **Sere (Seral Community)**: The entire sequential progression of transitional communities that successively replace one another in a given habitat: \text{Pioneer Community} \longrightarrow \text{Transitional Seral Stages} \longrightarrow \text{Climax Community}
- **Pioneer Species**: The first hardy organisms to colonize a previously uninhabited or barren area (e.g., **Crustose Lichens** and mosses on bare rock; **Phytoplankton** in barren water).
- **Climax Community**: The final, stable, mature, and self-perpetuating terminal community in dynamic equilibrium with the regional climate (e.g., a mature forest).
- **Primary Succession vs Secondary Succession**:

    - *Primary Succession*: Initiates on newly exposed barren surfaces where no previous soil or life existed (bare volcanic rock, cooled lava, newly formed sand dune).
    - *Secondary Succession*: Occurs on previously vegetated lands where pre-existing communities were destroyed by disturbance (forest fires, abandoned agricultural fields, floods), having pre-existing soil; proceeds substantially faster than primary succession.
- **Hydrosere vs Xerosere**:
    - *Hydrosere*: Succession sequence starting in an open body of fresh water (Phytoplankton \to Submerged macrophyte \to Floating \to Reed-swamp \to Marsh-meadow \to Woodland \to Forest).
    - *Xerosere (Lithosere)*: Succession sequence starting on dry, bare rock (Crustose lichen \to Foliose lichen \to Moss \to Herbaceous \to Shrub \to Forest).
- **Biogeochemical Cycles (Nutrient Cycling)**:
- Circulation of chemical elements (carbon, nitrogen, phosphorus, sulphur, water) through biotic and abiotic components.
- *Gaseous Cycles*: Reservoir in atmosphere or hydrosphere (e.g., Nitrogen, Carbon, Oxygen).
- *Sedimentary Cycles*: Reservoir in Earth's crust / lithosphere rocks (e.g., Phosphorus, Sulphur).

2.14 Ecotones, Edge Effect, Bioindicators & Biofertilizers

- **Ecotone & Edge Dynamics**:
- **Ecotone**: The transitional boundary or junction zone between two distinct, adjoining ecosystems or biomes (e.g., **Mangroves** between marine and terrestrial ecosystems; **Estuaries** between river and sea; **Marshes** between aquatic and dry land).
- **Edge Effect**: The ecological phenomenon where the boundary ecotone zone exhibits a significantly higher species richness, population density, and diversity than either of the adjacent communities alone.
- **Edge Species**: Organisms primarily or most abundantly found in the ecotone boundary zone (e.g., certain warblers, edge-nesting birds).
- **Ecocline**: A physiochemical or environmental gradient (such as elevation, moisture, salinity, or temperature) along which species varieties and community structures gradually shift.
- **Ecological Bioindicators**:
- **Lichens**: Dual organisms consisting of symbiotic fungus and alga/cyanobacterium. Highly sensitive bioindicators of atmospheric **Sulphur Dioxide (SO_2)** pollution and heavy metals. They completely disappear ("lichen desert") in polluted urban and industrial atmospheres.
- **Mosses & Algae**: Sensitive to acid precipitation and heavy metals.
- **Aquatic Macrophytes & Biofertilizers in Agriculture**:
- **Azolla**: A free-floating aquatic freshwater fern that harbors the symbiotic cyanobacterium ***Anabaena azollae*** in its dorsal leaf cavities. Widely inoculated in flooded paddy (rice) fields as a potent green biofertilizer, fixing 30-50\text{ kg N/ha}.
- **Other Aquatic Macrophytes**:
    - *Lemna* (Duckweed): Fast-growing floating macrophyte used in wastewater phytoremediation.

    - *Wolffia*: Rootless, microscopic aquatic plant (smallest flowering plant on Earth).
    - *Trapa* (Water Chestnut / Singhara): Rooted floating aquatic macrophyte cultivated in freshwater ponds for edible nutritious kernels.

- **Pioneer Concept Milestones (Direct Exam Locks)**:
- **Raymond Lindeman (1942)**: Formulated the **10% Law of Trophic Efficiency** (only \approx 10\% of energy stored as biomass in one trophic level is transferred to the next higher level; 90\% is lost via respiration and metabolic heat).
- **John W. Bennett (1976)**: Introduced the concept of **'Ecological Transition'** (*The Ecological Transition: Cultural Anthropology and Human Adaptation*).
